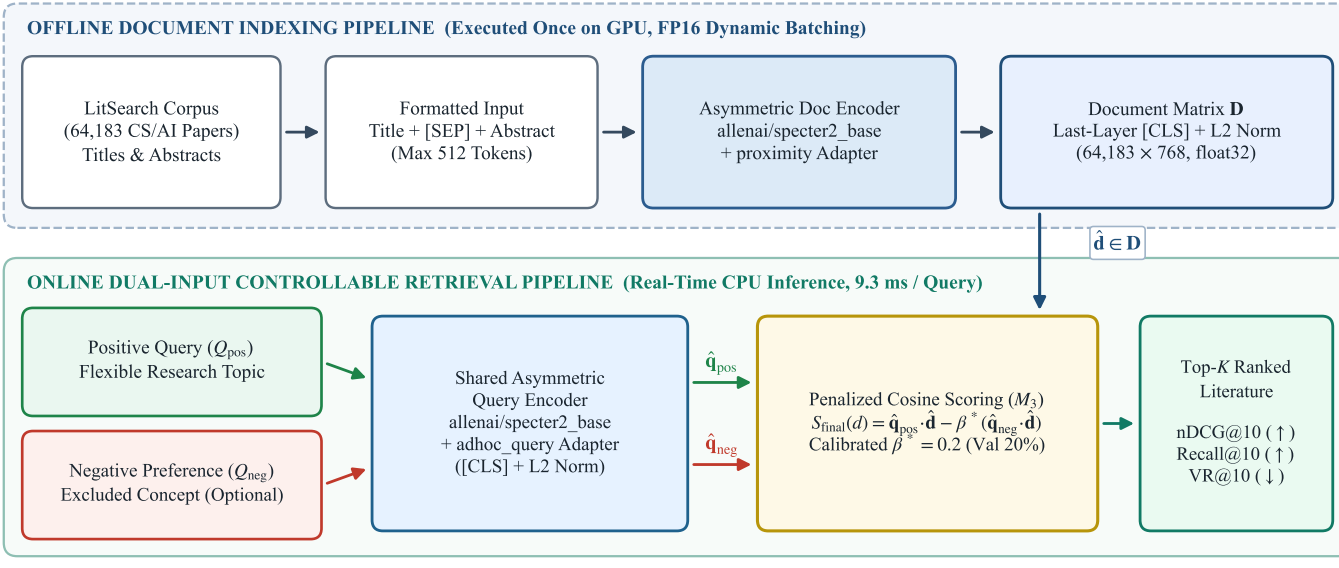

OFFLINE DOCUMENT INDEXING PIPELINE (Executed Once on GPU, FP16 Dynamic Batching)
LitSearch Corpus
(64,183 CS/AI Papers)
Titles & Abstracts
Formatted Input
Title + [SEP] + Abstract
(Max 512 Tokens)
Asymmetric Doc Encoder
allenai/specter2_base
+ proximity Adapter
Document Matrix D
Last-Layer [CLS] + L2 Norm
(64,183 × 768, float32)
$\hat{\mathbf{d}} \in \mathbf{D}$
ONLINE DUAL-INPUT CONTROLLABLE RETRIEVAL PIPELINE (Real-Time CPU Inference, 9.3 ms / Query)
Positive Query (Q_{pos})
Flexible Research Topic
Negative Preference (Q_{neg})
Excluded Concept (Optional)
Shared Asymmetric
Query Encoder
allenai/specter2_base
+ adhoc_query Adapter
([CLS] + L2 Norm)
$\hat{\mathbf{q}}_{pos}$
$\hat{\mathbf{q}}_{neg}$
Penalized Cosine Scoring (M_3)
$S_{final}(d) = \hat{\mathbf{q}}_{pos} \cdot \hat{\mathbf{d}} - \beta^* (\hat{\mathbf{q}}_{neg} \cdot \hat{\mathbf{d}})$
Calibrated $\beta^* = 0.2$ (Val 20%)
Top-K Ranked
Literature
nDCG@10 (↑)
Recall@10 (↑)
VR@10 (↓)